26/2/2026

Num. Integration : To find $\int_a^b f(x)dx$, where $y = f(x)$

Recall

① Trapezoidal Rule : 'n' → no restrictions

simple formula $\int_a^b f(x)\,dx = \frac{h}{2}\left[(y_0+y_n)+2(y_1+y_2+\cdots+y_{n-1})\right]$

② Simpson's $\frac{1}{3}$rd rule : 'n' even

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[(y_0+y_n)+2(y_2+\underbrace{y_4+\cdots+y_{n-2}}_{\text{even}})+4(\underbrace{y_1+y_3+\cdots+y_{n-1}}_{\text{odd}})\right]$$

③ Simpson's $\frac{3}{8}$th rule : 'n' should be a multiple of 3

$$\int_a^b f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n)+2(y_3+y_6+\cdots+y_{n-3})+\underline{\underline{3}}(y_1+y_2+y_4+y_5+\cdots+y_{n-1})\right]$$

Q1:

$$\int_0^1 \frac{dx}{1+x^2}, \qquad n=6,\ h=\frac{1}{6}$$

Simpson's $\frac{3}{8}$th:

$$\therefore \int_0^1 \frac{dx}{1+x^2} = \frac{3\cdot 1}{6\times 8}\left[(y_0+y_6)+2(y_3)+3(y_1+y_2+y_4+y_5)\right]$$

$$= 0.7854$$

[Actual: $\tan^{-1}(1)-\tan^{-1}(0) = \frac{\pi}{4} \simeq \frac{3.14}{4} = 0.7854$

② $\int_0^{\pi} \sin x \, dx = 2.0019$

$n=6$

by Simpson's $\frac{3}{8}$th rule

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ form}$$

$$= \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

(by l'Hospital's rule)

3. E.V. $\int_0^{\pi/2} \frac{\sin x}{x}\,dx$ using Simpson's $\frac{3}{8}$th rule.

Sol$^{\underline{n}}$: Let $y = f(x) = \frac{\sin x}{x}$

Let's take $n = 6$

$\therefore h = \frac{\frac{\pi}{2} - 0}{6} = \frac{\pi}{12}$

$\lim_{x \to 0} \frac{\sin x}{x} \left(\frac{0}{0}\text{ form}\right) = \lim_{x \to 0} \frac{\cos x}{1}$

By Simpson's $\frac{3}{8}$th rule,

$$\int_a^b f(x)\,dx = \frac{3h}{8}\Big[(y_0 + y_n) + 2(y_3 + y_6 + \cdots + y_{n-3}) + 3(y_1 + y_2 + y_4 + \cdots + y_{n-1})\Big]$$

Here, $\int_0^{\pi/2} \frac{\sin x}{x}\,dx = \frac{3h}{8}\left[(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)\right]$

Table

x	$x_0 = 0$	$x_1 = \pi/12$	$x_2 = \frac{2\pi}{12}$	$x_3 = \frac{3\pi}{12}$	$\frac{\pi}{12}$ $\frac{2\pi}{12}$ $x_4 = 4\pi/12$
$y = \frac{\sin x}{x}$	$y_0 = \frac{\sin 0}{0} = 1$	y_1 $\frac{\sin \pi/12}{\pi/12} = 0.9886$	y_2 0.9543	y_3 0.9003	y_4 0.8270
x	$x_5 = \frac{5\pi}{12}$	$x_6 = \frac{\pi}{12}$			
y	y_5 0.7379	y_6 0.6366			

$\therefore \int_0^{\pi/2} \frac{\sin x}{x}\,dx = \frac{3 \times \pi}{12 \times 8}\Big[\qquad\Big] = 1.3708$